

Steps of CART Algorithm:

1. **Start with the Root Node:** The entire dataset is at the root.
2. **Choose the Best Feature to Split:**
 - For classification, CART uses **Gini Impurity** (or sometimes Entropy) to determine the best split.
 - For regression, CART uses **Mean Squared Error (MSE)** to minimize variance.
3. **Recursively Split Data:** The dataset is split into left and right child nodes.
4. **Continue Splitting Until a Stopping Condition is Met:**
 - A predefined **maximum depth** is reached.
 - The **minimum number of samples per node** is too small.
 - There is **no further significant reduction in impurity**.
5. **Pruning:** After building a full tree, CART often prunes branches to reduce overfitting.

Shannon Entropy

Shannon Entropy is a measure of **uncertainty** in a dataset. It quantifies how **impure** or **disordered** the data is. The formula for entropy is:

$$H(X) = - \sum p_i \log_2 p_i$$

Information Gain (H Information)

Information Gain (IG) measures **how much entropy is reduced** after a split. It tells us **how valuable** a feature is for splitting the data.

$$IG = H(\text{Parent}) - \sum \left(\frac{n_{\text{child}}}{n_{\text{total}}} \times H(\text{Child}) \right)$$

- A **higher Information Gain** means the split reduces uncertainty significantly.
- The best split is the one that **maximizes Information Gain**.

Gini Impurity

Gini Impurity is a measure of how often a randomly chosen element from the dataset would be **incorrectly classified** if it were randomly labeled based on the class distribution. It quantifies the **impurity** of a dataset, where lower values indicate purer splits.

The formula for Gini Impurity is:

$$G = 1 - \sum p_i^2$$

- **Gini = 0** → The dataset is **pure** (all instances belong to one class).
- **Gini = 0.5** → Maximum impurity (equal probability of each class).

Cost Complexity Pruning (cp) is a **regularization parameter** used in **CART (Classification and Regression Trees)** to control tree size and prevent overfitting. It penalizes trees with too many splits by adding a complexity cost to the error function.

The cost complexity function is:

$$R_\alpha(T) = R(T) + \alpha \cdot |T|$$

- $R(T)$ = **Total misclassification error** (for classification) or **total variance** (for regression).
- $|T|$ = **Number of terminal (leaf) nodes** (tree complexity).
- α (**Complexity Parameter, cp**) = Controls the trade-off between tree size and accuracy.
- **Higher cp (e.g., cp = 0.05) → Smaller tree (more pruning)**
 - The model removes less significant splits.
 - Helps prevent **overfitting**.
- **Lower cp (e.g., cp = 0.001) → Larger tree (less pruning)**
 - Allows more splits, capturing more details in the training data.
 - Can lead to **overfitting** if too small.